Machine Learning Project
Alfatraining

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1 Project Description

The increasing prevalence of sustainable mobility concepts, bike-sharing systems are gaining in importance in urban areas especially in large cities. For operators of such systems, however, it is crucial to be able to predict bicycle demand as accurately as possible in order to plan availability efficiently and allocate resources optimally.
This project develops a machine learning model that predicts the number of rented bicycles based on historical data. The analysis is based on the London Bike Sharing Dataset, which, in addition to information on the number of bike rentals, also contains various influencing factors such as weather conditions, time variables, and seasonal effects. The dataset can be found on
https://www.kaggle.com/datasets/hmavrodiev/london-bike-sharing-dataset
By applying various machine learning algorithms, the project aims to investigate which factors have the strongest influence on bicycle use and which model provides the most accurate predictions.

1.1 The aim of the investigation

The goal of this study is to develop a machine learning model that can predict the number of rented bicycles as accurately as possible. In particular, the following questions will be addressed:

What factors influence the demand for bicycles?

What role do weather conditions, time of day, and seasonal effects play?

Which machine learning method is best suited for predicting bicycle usage?

The results of the analysis can contribute to a better understanding of the dynamics of urban bicycle sharing systems and improve bicycle availability planning.

1.2 Working hypothesis

Before the analysis begins, it is assumed that the demand for bicycles is significantly influenced by environmental and time factors. In particular, the following variables are expected to have a strong influence on the target variable:

Temperature: Higher temperatures are likely to lead to increase in bicycle use.

Humidity: High humidity could reduce bicycle use.

Weather conditions: Adverse weather conditions such as rain or fog could decrease demand.

Time of day: Increased use is expected during peak commuting hours in the morning and evening.

Weekends and holidays: Leisure activities could influence demand on these days.

Furthermore, non-linear relationships between the variables are suspected, which is why more complex machine learning models may deliver better results than simple linear models.

1.3 Roadmap of Investigation Process

This machine learning project is being carried out in several systematic steps. The goal of this approach is to develop the most robust model possible for predicting the number of rented bicycles while simultaneously avoiding overfitting.

1. The first step involves conducting an exploratory data analysis (EDA) to gain a fundamental understanding of the data. This includes calculating statistical measures, analyzing distributions, and investigating potential relationships between the variables.

2. Data cleaning and preparation, this includes handling missing values, verifying data types, and feature engineering.

3. In the next step, the data is split into training and test datasets. The training dataset is used for training and optimizing the machine learning models, while the test dataset is used exclusively for the final evaluation of model performance.

4. Various machine learning models are trained. For each model, suitable hyper-parameters are optimized using cross-validation.

5. The model with the optimal hyper-parameters has been determined is applied to the previously retained test dataset to obtain an objective evaluation of the predictive performance.

1.4 Quality criteria (score)

Various regression metrics are used to assess model quality. This includes *Mean Absolute Error (MAE)*, *Root Mean Squared Error (RMSE)* and *mean absolute percentage error (MAPE)*. The aim is to an accuracy score with 94% or higher on the test data set.

2 Data Analysis

The dataset frame have $(17414, 10)$ shape with the following features :

- timestamp : timestamp field for grouping the data
- cnt : the count of a new bike shares
- t1 : real temperature in C
- t2 : temperature in C "feels like"
- hum : humidity in percentage
- wind _speed : wind speed in km/h
- weather _code : category of the weather
- is _holiday : boolean field , 1 holiday / 0 non holiday
- is _weekend : boolean field , 1 if the day is weekend
- season : category field meteorological seasons with 0 spring , 1 summer , 2 fall , 3 winter.
- weathe _code category description: 1 = Clear , mostly clear but have some values with haze/fog/patches of fog/ fog in vicinity 2 = scattered clouds / few clouds 3 = Broken clouds 4 = Cloudy 7 = Rain/ light Rain shower/ Light rain 10 = rain with thunderstorm 26 = snowfall 94 = Freezing Fog

The central feature or the target column is the count feature. To analyze the relationship between data set features and the target value one can plot the correlation values. The correlation analysis provides insight into how the explanatory variables relate to the target variable bike share count (cnt). As shown in the correlation Fig. (2.1), the strongest positive relationships with bike demand are observed for the temperature variables: t1 (correlation ≈ 0.39) and t2 (correlation ≈ 0.37). This indicates that higher temperatures are associated with increased bike usage, which is consistent with expected seasonal mobility patterns. Wind speed shows only a weak positive correlation (≈ 0.12), suggesting it has limited influence on demand. Several variables exhibit weak negative correlations with the target, including holiday (≈ -0.05), weekend (≈ -0.10), season (≈ 0.12), and weather code (≈ -0.17), implying slightly lower bike usage under these conditions. The most notable negative relationship is observed with humidity (≈ -0.46), indicating that higher humidity levels tend to reduce bike sharing activity. Overall, the correlation structure suggests that weather related variables, particularly temperature and humidity, are the most influential predictors of bike demand, while calendar related variables contribute comparatively less explanatory power.

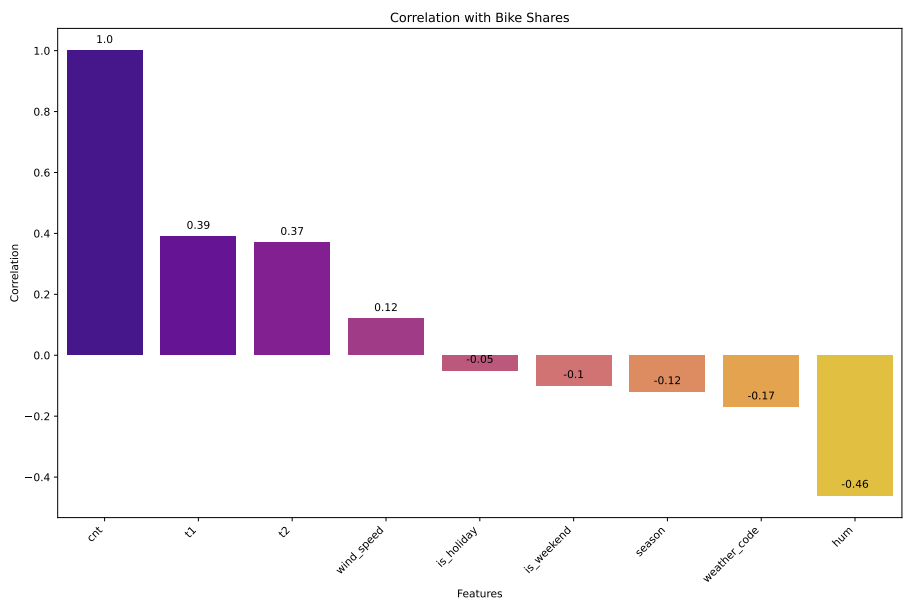

Figure 2.1: Correlation of Bikes rental counts vs. the other features

Other temporal trends can be extracted from the data set as illustrated in the figure below. For example, a clear jump in demand is noticed every day around 8 : 00 and 16 : 00 clock during work peak times. The same trend persist independent of season or being work day or a holiday.However, plotting the bike

rental count over the whole time period shows a few outliers which needs to be processed before training.

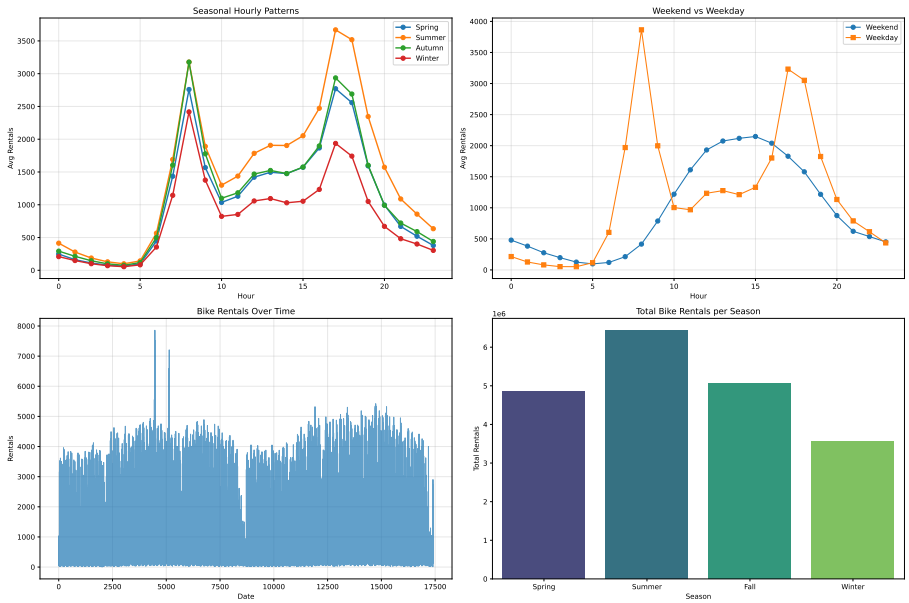

Figure 2.2: Different temporal tendencies of bikes share demand

3 Data Preprocessing

In order to prepare the data set for training we apply the following procedures :

1. convert the (timestamp) feature to datetime type using (pd.to_datetime()) method.
2. normalize datetime feature into year, month, day and hour then drop the timestamp column.
3. apply standard scaler transformation.
4. since the temperature and feeler temperature (t1,t2) are highly correlated it make sense to drop t2 and keep only t1 column.
5. outliers can skew the distribution of the target column, we use the IQR method to detect outliers and drop these data points from the data set. Fig.(3.1) shows that these outliers corresponds to two specific dates where a strike in public transportation, hence the high increase in demand for rntal bikes.

6. In addition we drop a data point with zero rental bikes which turnout to be the time when summer time zone starts in UK.

7. finally, we apply a train _test split to the data set with 0.8 vs 0.2 ratio.

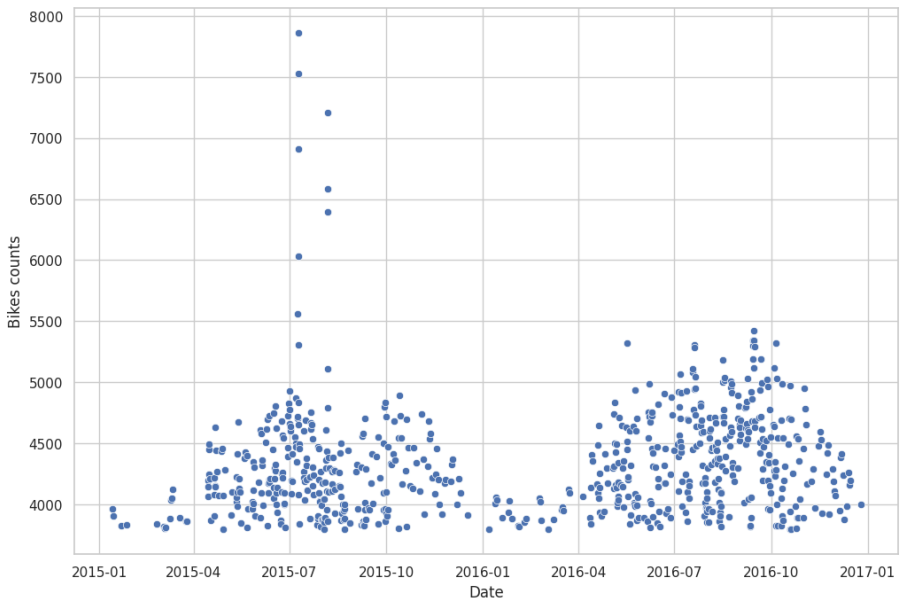

Figure 3.1: distribution of outliers over the time period

Note: the group tried other transformations too like applying log scale to the target column, use one hot encoder on weather feature, split the time features into its sin and cos components to explore periodicity in the time series. However, no significant effect is observed in the performance of models.

4 ML Models and scores

Our group used the training data to train the following models:

No.	Algorithm	Advantages	Disadvantages
1	MLP Regressor (Neural Network)	• Can learn very complex non-linear patterns • Highly flexible architecture with multiple hidden layers • Often produces high prediction accuracy	• Training can be slow • Requires tuning many hyperparameters • Hard to interpret (black-box model)
2	Random Forest	• Combines many decision trees for more stable predictions • Reduces overfitting compared to a single tree • Handles non linear relationships well • Provides feature importance	• Requires more computational resources • Model is harder to interpret • Large models can consume more memory
3	Gradient Boosting	• Often achieves very high prediction accuracy • Models learn sequentially from previous errors • Very effective for complex structured datasets	• Training time can be slow • Requires careful hyperparameter tuning • Can overfit if too many boosting iterations are used

Table 4.1: Advantages and disadvantages of selected machine learning algorithms

Using a grid search and cross validation, one can obtain the appropriate parameters to optimize the models. Model performance on the test dataset was evaluated using Mean Absolute Percentage Error (MAPE).

Models hyberparameters and MAPE score		
Gradient Boosting	Randomforest	MLP regressor
learning _rate:0.2	max _features: sqrt	hidden _layer _sizes : (20,20)
max _depth:5	max _depth: 30	alpha: 0.1
n _estimators:200	n _estimators:300	learning rate: 0.001
others : defaults	others : defaults	others : defaults
MAPE: 2.84 %	MAPE: 3 %	MAPE: 2.8 %

5 Final Coclusion

The analysis shows that bike demand is primarily driven by time-of-day patterns, while weather conditions and calendar events act as secondary factors.

6 Lessons learned

- There is no teamwork without a team leader !!
- Always try different scoring functions.
- There is no unique solution in ML projects